

1. Frobenius manifold.

defn A Frobenius algebra consists of $(A, \circ, e, \eta)$ where:

• $(A, \circ, e)$ is a unital associative commutative algebra.

• η is symmetric nondegenerate bilinear form. s.t. $\eta(X \circ Y, Z) = \eta(X, Y \circ Z)$

rmk. A Frobenius algebra is equivalent to a 2D TFT

• What is a Frobenius manifold?

A family of Frobenius algebras (2D TFT) lying on tangent spaces of a manifold that satisfy some conditions.

• Why Frobenius manifolds?

• They have appeared in a wide range of contexts:

1. quantum cohomology of a projective variety.

where $M = H^*(X; \mathbb{C})$

2. Calabi-Yau geometries. where $M = H^*(X; \mathbb{C})$ (S. Barannikov, M. Kontsevich)

3. singularity theory, where $M = M_f$

4. invariant theory of Coxeter groups $M = \mathcal{V}/\mathcal{W}$

5. families of curves (sometimes a F mfd).

1) on the base of spectral curve of some integrable system

2) on some Hurwitz space where the families of curves

are given by the universal curve. $L(p, q) \quad L = [L, 13]$

$$\sum_{i,j} = Z(\det(\lambda I - L))$$

6. space of isomonodromy deformations.

- Some Frobenius mflds from different contexts turn out to be the same.

- A concrete definition of a Frobenius mfld:

Defn. A Frobenius mfld is a tuple $M = (M, \circ, \eta, e, E)$ where:

- M is a complex mfld.

$$\eta \in \mathcal{P}(M, T^*M \otimes T^*M), \quad e, E \in \mathcal{P}(M, TM).$$

- each $p \in M$, $(T_p M, \circ, e, \eta)$ is a Frobenius algebra.

with the following conditions.

(i) η is flat, denote ∇ to be the Levi-Civita connection

(ii) e is horizontal : $\nabla e = 0$

(iii) let $c(X, Y, Z) := \eta(X \circ Y, Z)$ then

$\nabla_w c(X, Y, Z)$ is totally symmetric in w, X, Y, Z

(iv) $\nabla(\nabla E) = 0$ and $\mathcal{L}_E \circ = 0$, $\mathcal{L}_E \eta = (2-d)\eta$

(where $d \in \mathbb{Z}$ is called charge).

what does the definition tell you locally:

(i) local existence of flat coordinates t_i

$$\text{s.t. } \eta(t_i, t_j) = \delta_{ij}$$

(ii) we may assume $e = \frac{\partial}{\partial t_0}$

(iii) choose t_i as in (i) (ii). then local existence of

"pre-potential" $F(t)$, s.t.:

$$\eta(X, Y, Z) = X Y Z (F)$$

$$\text{in particular: } \eta_{ij} = \eta\left(\frac{\partial}{\partial t_i}, \frac{\partial}{\partial t_j}\right) = \frac{\partial^3 F}{\partial t_0 \partial t_i \partial t_j}$$

(iv) F gives a grading in t_i such that F is quasi-homogenous.

note that $\nabla E \in T(M, TM \otimes T^*M) = \text{End}_{\mathcal{O}_M} TM$

is flat.

we assume ∇E is diagonalized: $\nabla_{\partial_i} E = (1 - q_i) \partial_i$

$$\nabla_{\partial_i} E = \nabla_{\partial_i} (E^e \partial_e) = (\partial_i E^e) \partial_e \Rightarrow \partial_i E^e = \delta_{ie} (1 - q_i)$$

$$(\mathcal{L}_E \eta)_{\alpha\beta} = (\partial_\alpha E^e) \eta_{e\beta} + (\partial_\beta E^e) \eta_{\alpha e}$$

$$= [(1 - q_\alpha) + (1 - q_\beta)] \eta_{\alpha\beta} = (2-d) \eta_{\alpha\beta}$$

$$\Rightarrow (q_\alpha + q_\beta - d) \eta_{\alpha\beta} = 0 \quad (*)$$

$$(\mathcal{L}_E \eta)_{\alpha\beta}^r = E^e \partial_e C_{\alpha\beta}^r - \partial_e E^r C_{\alpha\beta}^e + \partial_\alpha E^e C_{e\beta}^r + \partial_\beta E^e C_{\alpha e}^r = C_{\alpha\beta}^r$$

$$\Rightarrow E^e \partial_e C_{\alpha\beta}^r = (q_\alpha + q_\beta - q_r) C_{\alpha\beta}^r \quad (**)$$

$$\begin{aligned}
\Rightarrow E^\epsilon \partial_\epsilon L_{\alpha\beta\gamma} &= E^\epsilon \partial_\epsilon (C_{\alpha\beta}^{\gamma'} \eta_{\gamma'\gamma}) \\
&= E^\epsilon \partial_\epsilon (C_{\alpha\beta}^{\gamma'} \eta_{\gamma'\gamma}) \\
&= (\eta_\alpha + \eta_\beta - \eta_\gamma) C_{\alpha\beta}^{\gamma'} \eta_{\gamma'\gamma} \\
&= (\eta_\alpha + \eta_\beta) C_{\alpha\beta\gamma} - C_{\alpha\beta}^{\gamma'} (\eta_{\gamma'} \eta_{\gamma'\gamma}) \\
&= (\eta_\alpha + \eta_\beta + \eta_\gamma - d) C_{\alpha\beta\gamma}
\end{aligned}$$

triple integral:

$$\mathcal{L}_E(F) = E^\epsilon \partial_\epsilon F = (3-d)F + \text{quadratic}$$

ymk associativity $\Rightarrow$ WDVV Equation for F :

$$(\partial_{\alpha\beta\gamma}^3 F) \eta^{\gamma\delta} (\partial_{\delta}^3 \eta_{\mu\nu} F) = \alpha \longleftrightarrow \mu$$

2. Saito Frobenius Mod.

- let $f: (\mathbb{C}^n, 0) \rightarrow (\mathbb{C}, 0)$ be an isolated singularity.
then we can get a miniversal deformation (universal unfolding)
of f of the form $F(x, \xi) = f + \sum_{i=1}^n s_i g_i$ where $g_1, \dots, g_n$ generate
the local algebra $\mathcal{O}_{\mathbb{C}^n, 0} / \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right) = \mathcal{T}_f$

- $\Delta = B_S' \subseteq \mathbb{C}$

$$S = B_1^n \subseteq \mathbb{C}^n$$

$$\mathcal{X} = F^{-1}(\Delta) \cap (B_2^n \times \mathbb{C})$$

$p: \mathcal{X} \rightarrow S$ be the projection.

2.1 product structure on T_S

Observation $\mathcal{T}_f = \dim S$ and is equipped with an algebra structure

Theorem the following $\mathcal{O}_S$ -homomorphism

$$\rho: T_S \rightarrow p_* \mathcal{O}_{\mathcal{X}} = p_* \left(\mathcal{O}_{\mathcal{X}} / \left(\frac{\partial F}{\partial x_1}, \dots, \frac{\partial F}{\partial x_n} \right) \right)$$

$$s \mapsto \hat{s} F|_p$$

for example

$$\left(\frac{\partial}{\partial s_i} \mapsto \left[\frac{\partial F}{\partial s_i} \right] \right)$$

where $\hat{s}$ is a lifting of s on $\mathcal{X}$

relative to p , is an isomorphism, called

the Kodaira-Spencer map.

"Proof" : 1. at $S=0$: by construction

2. show $p_* \mathcal{O}_C$ is locally free
this comes from p is finite and flat.

Def (Product on $\mathcal{T}_S$)

for $s, s' \in \mathcal{P}(S, \mathcal{T}_S)$. we define $s \circ s'$ by:

$$\widehat{s \circ s'} F := \hat{s} F|_C \circ \hat{s}' F|_C$$

Ex (A_n) for $F = x^{n+1} + s_0 + \dots + s_{n-1} x^{n-1}$

$$\partial_i \circ \partial_j \Big|_{S=0} = \begin{cases} \partial_{i+j} & \text{if } i+j \leq n \\ \partial_{i+j-n} & \text{if } i+j > n \end{cases}$$

2.2 unit and Euler vector field.

Def the unit $e \in \mathcal{P}(S, \mathcal{T}_S)$ is defined by

$$KS^{-1}(1), \text{ if we choose } g_i = 1, \text{ then } e = \frac{\partial}{\partial s_i}$$

Def the Euler vector field $E \in \mathcal{P}(S, \mathcal{T}_S)$ is defined by

$KS^{-1}(F)$, that is, the unique vector field

satisfying $[E(F)] = [F]$ in $p_* \mathcal{O}_C$

remark if we can give weights d_i to x_i , and r_i to s_i

such that: $F(x, s)$ is quasi-homogeneous of

total weight N , then $E = \frac{1}{N} \sum_{i=1}^n r_i s_i \frac{\partial}{\partial s_i}$ since we

have Euler identity:

$$NF = \left(\sum r_i s_i \frac{\partial}{\partial s_i} + \sum d_i x_i \frac{\partial}{\partial x_i} \right) F$$

$$\equiv \left(\sum r_i s_i \frac{\partial}{\partial s_i} \right) F \quad \text{mod } \frac{\partial F}{\partial x_1}, \dots, \frac{\partial F}{\partial x_n}$$

eg. (h_n) for $F = x^{n+1} + s_0 + \dots + s_{n-1} x^{n-1}$

$$E = \frac{1}{n+1} \sum_{i=0}^{n-1} (n+1-i) s_i \frac{\partial}{\partial s_i}$$

To give a suitable "metric" η we need to introduce:

2.3 Metric on $\tilde{T}_S$

Def The sheaf of relative holomorphic differential i -forms is defined as

$$\Omega_F^i = \Omega^i \mathcal{X}/S / \text{im} \left(\Omega^{i-1} \mathcal{X}/S \rightarrow \Omega^i \mathcal{X}/S \right)$$

Observation: Ω_F^n is a free $\mathcal{O}_C$ sheaf of rank 1

Proof: $\mathcal{X} = U \times S : \Omega^i \mathcal{X}/S \cong \sum_{n_1 < n_2 < \dots < n_i} \mathcal{O}_{\mathcal{X}} [dx_{n_1} \wedge \dots \wedge dx_{n_i}]$

in particular $\Omega^{n-1} \mathcal{X}/S \xrightarrow{\wedge dF} \Omega^n \mathcal{X}/S$ is given by:

$$\sum_{k=1}^n \mathcal{O}_{\mathcal{X}} [dx_1 \wedge \dots \wedge \hat{dx}_k \wedge \dots \wedge dx_n] \cong \mathcal{O}_{\mathcal{X}}^{\oplus n} \rightarrow \mathcal{O}_{\mathcal{X}} : (g_1, \dots, g_n) \mapsto \sum g_i \frac{\partial F}{\partial x_i}$$

There is a natural pairing on Ω_F^n defined by

def Grothendieck Residue

for $w_1, w_2 \in \Omega_F^n$ define:

$$J(w_1, w_2)/S = \text{Res} \left(\frac{\psi_1 \psi_2}{\frac{\partial F}{\partial x_1} \dots \frac{\partial F}{\partial x_n}} \right) = \left(\frac{1}{2\pi i} \right)^n \oint_{\gamma} \frac{\psi_1 \psi_2}{\frac{\partial F}{\partial x_1} \dots \frac{\partial F}{\partial x_n}} dz$$

where γ is a real n d'l cycle containing $\{ |\frac{\partial F}{\partial x_i}| = \epsilon \}$

with orientation fixed by $d(\arg F_1) \wedge \dots \wedge d(\arg F_n)$

Fact J is nondegenerate, and symmetric.

Therefore, a metric on $p_1 \mathcal{O}_C$ (therefore T_S) can be defined.

after fixing a generator ζ of $p_1 \Omega^n_{X/S}$:

$$\eta(f_1, f_2) := J(f_1 \zeta, f_2 \zeta)$$

A suitable choice of a generator is called a primitive form
which is studied by Kyoji Saito

Theorem (K. Saito, M. Saito)

for any isolated singularity.
there exists a primitive form ζ such that
under the isomorphism $J_S \cong p_1 \mathcal{O}_C \cong p_1 \Omega^n_F$

$(S, 0, e, E, \eta)$ is a Frobenius manifold

Example for simple singularities $f: \mathbb{C}^3 \rightarrow \mathbb{C}$

$$\zeta = [dx \wedge dy \wedge dz]$$

rmk. ζ is not unique (even up to scaling)

for simply elliptic singularities, the spaces
of primitive form is 1 d'l.

3. Examples

$$3.1 \quad F = -\frac{1}{3} z^3 + s_2 z + s_1$$

$$\frac{\partial F}{\partial z} = -z^2 + s_2$$

$$KS\left(\frac{\partial}{\partial s_1}\right) = 1$$

$$KS\left(\frac{\partial}{\partial s_2}\right) = [z]$$

$$\eta\left(\frac{\partial}{\partial s_i}, \frac{\partial}{\partial s_j}\right) = \text{Res}_{\frac{z}{s}} \left(KS\left(\frac{\partial}{\partial s_i}\right) \cdot KS\left(\frac{\partial}{\partial s_j}\right) [dz] \right)$$

$\therefore$ suffices to compute.

$$\frac{1}{2\pi i} \oint \frac{z^l}{z^2 - s_2} dz \quad \text{for } l = 0, 1, 2.$$

$\|z^2 - s_2\| < \epsilon$

$$\Rightarrow \eta\left(\frac{\partial}{\partial s_i}, \frac{\partial}{\partial s_j}\right) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$\therefore s_1, s_2$ gives flat coordinates

to compute the prepotential :

$$\frac{\partial^3 F}{\partial s_i \partial s_j \partial s_k} = \eta\left(\frac{\partial}{\partial s_i} \circ \frac{\partial}{\partial s_j}, \frac{\partial}{\partial s_k}\right)$$

similar computations give: another nonzero term:

$$\frac{\partial^3 F}{\partial s_2 \partial s_2 \partial s_2} = \frac{1}{2\pi\sqrt{-1}} \int \frac{z^3}{z^2 - s_2} \mathbb{1}_{\|z^2 - s_2\| < \epsilon}$$

$$= s_2$$

$$\therefore F = \frac{1}{2} s_1^2 s_2 + \frac{1}{4} s_2^4$$

3.2. $F = -\frac{1}{2}y^2 + x^2 + ax + b$

the same computations because

$$\frac{1}{2\pi\sqrt{-1}} \oint \frac{1}{\frac{\partial F}{\partial y}} dy = -1$$

3.3 cusp singularities.

Defn A cusp singularity is of the form

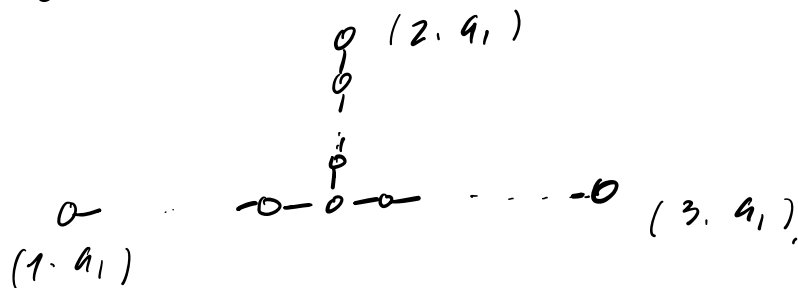
$$f_A = x_1^{q_1} + x_2^{q_2} + x_3^{q_3} - g^{-1} x_1 x_2 x_3 \quad \begin{array}{l} a_i \in \mathbb{Z}_{>0} \\ g \in \mathbb{C} \setminus \{0\} \end{array}$$

Take
$$F_A = x_1^{q_1} + x_2^{q_2} + x_3^{q_3} - s_{MA}^{-1} \cdot x_1 x_2 x_3 + s_1 + \sum_{i=1}^3 \sum_{j=1}^{q_i-1} s_{i,j} \cdot x_i^j$$

Fact 1 There exists a unique primitive form η_A satisfying

$$\eta_A|_{s=0} = [s_{MA}^{-1} dx_1 \wedge dx_2 \wedge dx_3]$$

Fact 2 the diagram of intersections between vanishing cycles is given by



Fact 3 (Milnor-Tseng, Rossi, Takahashi-Shiraishi)

$$M(F_A, \eta_A) \cong \mathbb{Q}H(\mathbb{P}_{a_1, a_2, a_3}^1)$$

where $\mathbb{P}_{a_1, a_2, a_3}^1$ is $\mathbb{P}^1$ with three orbifold

points $\left[\frac{\mathbb{C}}{\mathbb{Z}_{a_i}} \right]$

Appendix

An understanding: $\mathcal{O}_X / \frac{\partial F_S}{\partial x_1} \dots \frac{\partial F_S}{\partial x_n}$ defines the normal direction of the deformation space.

prop $\mu(f, 0) = \sum_{x \in \text{crit}(F_t)} \mu(F_t, x)$ in a small nbhd of t

proof $\Phi: U \times T \rightarrow \mathbb{C}^n, T, (u, t) \mapsto \left(\frac{\partial F}{\partial x_1}, \dots, \frac{\partial F}{\partial x_n}, t \right)$

finite over $(0, 0) \Rightarrow$ finite for small U

consider $\Phi_* \mathcal{O}_{U \times T}$ it's coherent, by finiteness

$\nu(t) = \nu(\Phi_* \mathcal{O}_{U \times T}, (0, t)) := \sum_{x \in \text{cr}(F_t)} \mu(F_t, x)$

it's sufficient to show, Φ is flat at $(0, 0)$, since for finite morphisms, flatness is open.

so we reduce to $\mathcal{O}_{U \times T, (0, 0)} = \mathbb{C}\{x_1, \dots, x_n, t_1, \dots, t_n\}$ which is a regular local ring

For regular local ring Φ flat $\Leftrightarrow f_1, \dots, f_n, t_1, \dots, t_n$ is regular

$\Leftrightarrow f_1, \dots, f_n$ is regular in $\mathbb{C}\{x_1, \dots, x_n\}$

$\Leftrightarrow \dim(f^{-1}(0), 0) = 0$

homomorphic case

cor $p: C \rightarrow S$ is finite and flat

and $p_* \mathcal{O}_C$ is a free $\mathcal{O}_S$ sheaf of rank n

References

1. From primitive forms to Frobenius mfd's
for the construction by K. Saito, A. Takahashi
2. Summary of the Theory of Primitive Forms.
for more details - by A. Matsuo
3. Introduction to Singularities and Deformations
for deformation theory involved. by G.M. Greuel
C. Lossen
E. Shustin